Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **Python/C++ tooling**, **visualization-oriented workflows**, and robust **validation pipelines** for high-reliability systems, cross-functional autonomy development, and rapid debugging loops.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving script reliability and backend observability across **Linux**-based systems.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **production stability** and accelerating issue triage across distributed services and cross-functional engineering workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, translating requirements into reusable APIs and UI components while improving release traceability and testability for high-velocity product iterations.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while maintaining quality through automated validation checks and structured defect-triage practices with stakeholders.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling rigorous behavior analysis and reproducible benchmark runs across safety-sensitive scenarios that demanded clear visual diagnostics and reliable debugging outputs.
- Produced technical summaries and debugging diagnostics that converted research findings into implementation plans, improving **defect isolation speed** and strengthening collaboration across ML, systems, and software engineering teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across enterprise environments.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** through automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving extensibility and reliability while coordinating integration constraints with engineering and operations teams.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through script-driven checks, reproducible environments, and robust rollback mechanisms.

Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash
Visualization & UI: React, JavaScript, TypeScript, Dashboards, Data Visualization
Backend & APIs: Flask, FastAPI, REST APIs, WebSockets, Microservices, ETL Pipelines
Cloud & Infrastructure: AWS (S3, Glue, ECS), Azure, Docker, Kubernetes, Linux/UNIX
Testing & Validation: Script Testing, Unit/Integration Testing, Troubleshooting, CI/CD, Agile/Scrum

Projects

CPU Scheduling Simulator | *C++, React.js, JavaScript*

- Implemented FCFS, SJF, SRT, RR, and MLQ scheduling algorithms with visualization and metrics tooling, reducing debugging time by **20%** while strengthening low-level **C++** reasoning and performance-focused engineering practices.

PiXMod | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI visual editing platform using **SAM** segmentation and **Stable Diffusion** generation, processing **1K+ assets** while improving visual inspection workflows, metadata indexing, and production reliability across image pipelines.

CaptionCraft | *Transformers, PyTorch, Flask*

- Benchmarked transformer vision-caption models across **10K images** and **100K captions**, improving caption accuracy by **15%** and creating repeatable evaluation workflows that supported model validation and analysis quality.